

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Cummins® electronic service tool, or equivalent.

Additional Service Items

- No additional service items required.

General Information

The Air Handling Performance Test is a diagnostic test used to evaluate the health of the air handling components. Throughout the test, engine speed and the position of air handling actuators are controlled to create conditions for the various diagnostics to run.

Components checked in the test:

- Intake Air Throttle
- EGR Valve
- Variable Geometry Turbocharger
- EGR Cooler
- Exhaust Gas Pressure Sensor
- Intake Manifold Pressure Sensor
- EGR Differential Pressure Sensor
- Plugged or leaking EGR differential pressure ports

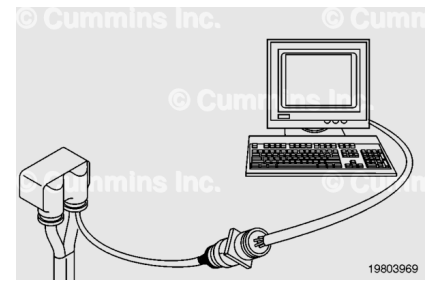
To start the test, engine **must** be at operating temperature and stopped with the keyswitch in the ON position. After initial actuator span checks, the test will prompt the user to start the engine. Once engine speed is detected, the test will automatically proceed with no further input from the user.

If an issue is detected, the issue **must** be addressed.

Run the test until it passes.

System Requirements

- ECM calibration code release date September 2021 or later.
- INSITE™ electronic service tool Version 8.7.0 or later.
- Cummins® Guidanz™ Mobile electronic service tool Version 6.2 or later.
- Cummins® Guidanz™ Windows® electronic service tool Version 2.6 or later.
- Check the engine control module (ECM) calibration revision history for calibration updates for this test. If the ECM does **not** contain that revision or higher, update the calibration. Refer to Procedure 019-032 in Section 19.
(/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)



Preparatory Steps

1. Select an appropriate location to park the unit.
2. Park the unit securely.
 - a. Set the parking brake.
 - b. Place the transmission in Park, if provided. Otherwise, place the transmission in Neutral.
 - c. Do **not** operate any power takeoff (PTO)-powered devices.
3. Verify the ambient conditions are met.
 - a. Ambient Air Pressure above 74 kPa [10.7 psi].
 - b. Ambient Air Temperature above -19°C [-2.2°F] and below 64°C [147.2°F].
4. Verify engine compartment is closed.
5. Warmup the engine. Use the recommended Cummins® electronic service tool or equivalent to verify the following:
 - a. Coolant Temperature above 70°C [158°F].
 - b. Intake Manifold Temperature above 20°C [68°F].
6. Verify aftertreatment temperatures are met.
 - a. Diesel Particulate Filter (DPF) Inlet Temperature below 250°C [482°F].
 - b. DPF Outlet Temperature below 250°C [482°F].

Note : If Fault Codes 1921 or 1922 are present, the soot load must be brought down to normal levels for this test to run due to an engine protection derate. After a regeneration, run the engine at high idle (greater than 1800 rpm) for 5 minutes.

Test

1. Begin the Air Handling Performance Test.

- Use the recommended Cummins® electronic service tool or equivalent to perform the test.
- Cycle the keyswitch. Select start button on test screen to begin.
- Start engine when prompted.

2. Monitor the area.

- Verify the vehicle and surrounding area are monitored during the test. If any unsafe condition occurs, shut the engine off immediately.
- To stop the test, select the STOP button on the test screen. Test can also be stopped by engaging the clutch, brake, throttle pedal or by turning the engine off.
- Test progress can be monitored in the status window.

At the conclusion of the test, engine will return to idle. The Air Handling Performance Test will display either a Pass, Fail or Abort message until the next key cycle.

If test passes, then no further action is required.

If the Air Handling Performance Test does **not** pass, reset the search to "Air Handling Performance Test Did **Not** Pass". **Only** select the solution title that matches the error message in the recommend Cummins® electronic service tool or equivalent.

If a repair is necessary, perform the repair and then run the Air Handling Performance Test until the test passes.

The Air Handling Performance Test will **not** start or will be aborted if:

- Preparatory conditions **not** within specification
- Accelerator pedal is depressed
- Clutch pedal is depressed
- Brake pedal is depressed
- Parking brake **not** set
- Parking brake configured incorrectly
- Transmission put into gear
- PTO engaged
- Vehicle speed detected
- Engine protection state active
- DPF Inlet Temperature greater than 350°C [662°F]
- DPF Outlet Temperature greater than 300°C [572°F].

If the Air Handling Performance Test will **not** start, reset the search to Air Handling Performance Test Will **Not** Start.

If the Air Handling Performance Test will **not** complete, reset the search to Air Handling Performance Test Will **Not** Complete.

If the test needs to be restarted, cycle the keyswitch OFF for 90 seconds and then back ON.

Finishing Steps

- Check for any active fault codes. If active fault codes are present, follow the appropriate troubleshooting.

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